Yoonjae Hwang

yoonjae@usc.edu | +82-10-2033-6031 | Seoul, Republic of Korea | yoonjaeh.github.io

EDUCATION

University of Southern California 08/2025 – 08/2026
Master of Science in Computer Science

University of Illinois at Urbana Champaign 08/2017 – 05/2023
Bachelor of Science in Materials Engineering; Minor in Computer Science

WORK EXPERIENCE

ASM 09/2026 – Present
Forward Deployed Engineer in Product Intelligence Team
- **Led end-to-end analysis of semiconductor deposition data**, applying statistical **modeling and machine learning** to generate Design of Experiments, Anomaly Detection, and Data Ingestion pipeline applications for product intelligence
- **Partnered directly with semiconductor chipmakers and vendors** to scope technical challenges, deliver demos and recommendations, and translate model results into business impact for both technical and executive audiences.

Ericsson 08/2023 – 06/2025
Software Engineer / Innovation Team Driver
- **Accelerated pre-production bug escapes by 25%** by architecting and deploying an intelligent, Kubernetes-native CI/CD pipeline, automating entire test and validation framework for the 5G Packet Core
- **Boosted end-to-end system test coverage by 40%** and cut manual QA cycle by developing 150+ functional and integration tests in Python and Erlang that auto-triaged failures, contributing to CI/ CD
- **Optimized core 5G mobility functions (AMF/SMF) by analyzing network traffic patterns** to identify and eliminate critical performance bottlenecks, resulting in a reduction in packet latency and a 6% increase in user session throughput

Samsung Electronics 07/2022 – 08/2022
Global Intern
- **Drove a projected increase in FEOL process yield** by **diagnosing** a critical, previously overlooked failure pattern in the redistribution layer through a rigorous root cause analysis of wafer defect data
- **Expedited partner integration** by translating complex semiconductor packaging protocols into a new repository of concise technical briefs, eliminating a key communication bottleneck between engineering and partner relations

RESEARCH EXPERIENCE

Robotics Embedded Systems Lab 02/2026 – 09/2026
University of Southern California
- **Distributed Multi-Robot Coordination:** Researching **Distributed Sequential Greedy Assignment (DSGA)** frameworks for online mapping and exploration, focusing on achieving computational **optimization in large-scale robot fleets**
- **Constrained Informative Planning:** Investigating **matroid-constrained submodular maximization** to coordinate autonomous agents in cluttered environments, integrating and ensuring "**sim-to-real**" consistency

Sim-to-Real Transfer for Robust Manipulation via Synthetic Data Generation 05/2026 – 08/2026
University of Southern California | Personal Research
- **Designing a sim-to-real transfer pipeline** for robotic manipulation using **ManiSkill3** (GPU-parallelized simulator), generating large-scale synthetic demonstration datasets via motion planning and reinforcement learning
- **Constraining robot action space** to 4-DOF (xyz translation + gripper) and trained policies using **PPO/SAC on a V100 GPU** cluster to **benchmark imitation learning (ACT/Diffusion Policy) policies**, achieving scalable data collection without physical hardware

Autonomous Networks Research Group 10/2025 – 08/2026
University of Southern California under Prof. Bhaskar Krishnamachari
- **Conceptualizing and benchmarking** novel task-scheduling algorithms to **minimize execution latency** in edge networks
- **Synthesizing a scalable, Python-based simulation framework** from the ground up to model and validate the performance of complex, DAG-based computational workflows across thousands of distributed nodes

High Energy and Dendrite Free Micro Batteries 08/2021 – 05/2022
Braun Group, Materials Science and Engineering, UIUC, United States
- **Pioneered** a high-energy micro-battery (300μm) using a silicon foam anode to overcome the volumetric expansion and low-density limitations of conventional materials, **achieving a 30% higher energy density** than the graphite-anode baseline
- **Demonstrated** breakthrough cell stability and suppressed dendrite formation by **maintaining 95% of initial capacity after charge/discharge cycles**, a performance metric validated through rigorous potentiostatic analysis

PROFESSIONAL SKILLS

Programming Languages/Tools
AWS, Bash, CI/CD, C++, Dart, Docker, Erlang, Firebase, Flutter, Git/GitHub, HTML/CSS, Java, JavaScript, Kubernetes, Python, PyTorch, React, REST API, SQL (MySQL, PostgreSQL)

Microfabrication and Microstructure Analysis
Oxidation, Photolithography, Etching, Cleanroom Experience, Glovebox Experience, OOF2, LAMMPS, Quantum Espresso, SEM, OM, EIS, Raman Spectroscopy, Potentiostat, DSC, LVDT

PUBLICATION

Y. Hwang and B. Krishnamachari, "Profiling and Scheduling Complex O-RAN Applications Across the 5G Edge and Cloud," Accepted to the International Conference on Computer Communications and Networks (ICCCN)

Y. Hwang, H. Kim, E. Novoseller, G. Shi and G. S. Sukhatme, "Failure-Resilient Multi-Robot Coordination with k-Connectivity Constrained Submodular Maximization," Submitted to the International Conference on Robotics and Automation (ICRA)